

MATH1560J Assignment 6 - Sample Answer

Exercise 6.1 (20 pts)

(i)

Use Integration by Parts on I_{n-1} . Let $u = \frac{1}{(1+x^2)^{n-1}} = (1+x^2)^{-(n-1)}$ and $dv = dx$.

Then $du = -(n-1)(1+x^2)^{-n} \cdot 2x dx = -2(n-1)\frac{x}{(1+x^2)^n} dx$ and $v = x$.

$$I_{n-1} = \int \frac{dx}{(1+x^2)^{n-1}}$$

$$I_{n-1} = uv - \int v du$$

$$I_{n-1} = \frac{x}{(1+x^2)^{n-1}} - \int x \left(-2(n-1)\frac{x}{(1+x^2)^n} \right) dx$$

$$I_{n-1} = \frac{x}{(1+x^2)^{n-1}} + 2(n-1) \int \frac{x^2}{(1+x^2)^n} dx$$

Now, rewrite x^2 in the integral: $x^2 = (1+x^2) - 1$.

$$I_{n-1} = \frac{x}{(1+x^2)^{n-1}} + 2(n-1) \int \frac{(1+x^2) - 1}{(1+x^2)^n} dx$$

$$I_{n-1} = \frac{x}{(1+x^2)^{n-1}} + 2(n-1) \left[\int \frac{1+x^2}{(1+x^2)^n} dx - \int \frac{1}{(1+x^2)^n} dx \right]$$

$$I_{n-1} = \frac{x}{(1+x^2)^{n-1}} + 2(n-1) \left[\int \frac{dx}{(1+x^2)^{n-1}} - \int \frac{dx}{(1+x^2)^n} \right]$$

$$I_{n-1} = \frac{x}{(1+x^2)^{n-1}} + 2(n-1) [I_{n-1} - I_n]$$

$$\text{Solve for } I_n : I_n = \frac{x}{2(n-1)(x^2+1)^{n-1}} + \frac{2n-3}{2(n-1)} I_{n-1} + C$$

(ii)

Let $J_n = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^n}$. The formula we need to prove is $J_{n+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi$.

Base Case: $n = 1$

$$J_2 = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2}$$

Using the reduction formula from part (i) for definite integrals:

$$J_2 = \left[\frac{x}{2(1)(x^2 + 1)^1} \right]_{-\infty}^{\infty} + \frac{2(2) - 3}{2(1)} J_1 = 0 + \frac{1}{2} J_1 = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{1 + x^2} = \frac{1}{2} [\arctan x]_{-\infty}^{\infty} = \frac{1}{2} \pi$$

The base case holds.

Inductive Step:

Assume the formula holds for some $k \geq 1$:

$$J_{k+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2k - 1)}{2 \cdot 4 \cdot 6 \cdots (2k)} \cdot \pi \quad (*)$$

We need to show the formula holds for $n = k + 1$, i.e.,

$$J_{k+2} = \frac{1 \cdot 3 \cdot 5 \cdots (2k - 1) \cdot (2k + 1)}{2 \cdot 4 \cdot 6 \cdots (2k) \cdot (2k + 2)} \cdot \pi$$

Using the reduction formula with $n = k + 1$:

$$J_{k+2} = \left[\frac{x}{2(k + 2 - 1)(x^2 + 1)^{k+2-1}} \right]_{-\infty}^{\infty} + \frac{2(k + 2) - 3}{2(k + 2 - 1)} J_{k+1} = \frac{2k + 1}{2k + 2} J_{k+1}$$

Substitute the inductive hypothesis (*):

$$J_{k+2} = \frac{2k + 1}{2k + 2} \cdot \left(\frac{1 \cdot 3 \cdot 5 \cdots (2k - 1)}{2 \cdot 4 \cdot 6 \cdots (2k)} \cdot \pi \right)$$

The induction holds.

Exercise 6.2 (30 pts)

(i)

Let $u = f(x)$ and $dv = dx$. Then $du = f'(x) dx$ and $v = x$.

$$\int u dv = uv - \int v du$$

$$\int f(x) dx = f(x) \cdot x - \int x \cdot f'(x) dx$$

(ii)

From part (i), we have the definite integral form:

$$\int_a^b f(x) dx = [xf(x)]_a^b - \int_a^b xf'(x) dx = bf(b) - af(a) - \int_a^b xf'(x) dx \quad (*)$$

For the remaining integral, $\int_a^b xf'(x) dx$, use the substitution rule.

Let $y = f(x)$. Since f and g are inverse functions, $x = g(y)$. $dy = f'(x) dx$.

When $x = a$, $y = f(a)$. When $x = b$, $y = f(b)$.

$$\int_a^b xf'(x) dx = \int_{f(a)}^{f(b)} g(y) dy$$

Substitute this back into (*):

$$\int_a^b f(x)dx = bf(b) - af(a) - \int_{f(a)}^{f(b)} g(y)dy$$

(iii)

Let $f(x) = \ln x$. The inverse function is $g(y) = e^y$. The limits are $a = 1$ and $b = e$.

Calculate the values of $f(x)$ at the limits: $f(a) = f(1) = \ln 1 = 0$, $f(b) = f(e) = \ln e = 1$

Using the formula from part (ii):

$$\int_1^e \ln x dx = e \cdot f(e) - 1 \cdot f(1) - \int_{f(1)}^{f(e)} g(y) dy = e \cdot 1 - 1 \cdot 0 - \int_0^1 e^y dy = e - e + 1 = 1$$

Exercise 6.3 (30 pts)

(i)

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$. Rewrite I_n :

$$I_n = \int_0^{\frac{\pi}{2}} \sin^{n-1} x \cdot \sin x dx$$

Let $u = \sin^{n-1} x$ and $dv = \sin x dx$. Then $du = (n-1) \sin^{n-2} x \cdot \cos x dx$ and $v = -\cos x$.

$$I_n = [-\cos x \sin^{n-1} x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x) \cdot (n-1) \sin^{n-2} x \cos x dx = 0 + (n-1) \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \sin^{n-2} x dx$$

$$I_n = (n-1) \left[\int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - \int_0^{\frac{\pi}{2}} \sin^n x dx \right] = (n-1)I_{n-2} - (n-1)I_n$$

$$\text{Solve for } I_n: nI_n = (n-1)I_{n-2} \iff \int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{n-1}{n} \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx$$

(ii)

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$. The formula is $I_{2n+1} = \frac{2}{3} \cdot \frac{4}{5} \cdots \frac{2n}{2n+1}$.

Base Case: $n = 1$

$$I_3 = \int_0^{\frac{\pi}{2}} \sin^3 x dx = \int_0^{\frac{\pi}{2}} \sin^2 x \sin x dx = \int_0^{\frac{\pi}{2}} (1 - \cos^2 x) d(-\cos x) = \int_1^0 (1 - u^2)(-du) = \frac{2}{3}$$

The base case holds.

Inductive Step:

Assume the formula holds for some $k \geq 1$:

$$I_{2k+1} = \prod_{j=1}^k \frac{2j}{2j+1} \quad (*)$$

We need to show the formula holds for $n = k + 1$, i.e.,

$$I_{2(k+1)+1} = I_{2k+3} = \prod_{j=1}^{k+1} \frac{2j}{2j+1} = \left(\prod_{j=1}^k \frac{2j}{2j+1} \right) \cdot \frac{2(k+1)}{2(k+1)+1}$$

Using the reduction formula from part (i) with $n = 2k + 3$:

$$I_{2k+3} = \frac{(2k+3)-1}{2k+3} I_{(2k+3)-2} = \frac{2k+2}{2k+3} I_{2k+1}$$

Substitute the inductive hypothesis (*):

$$I_{2k+3} = \frac{2k+2}{2k+3} \cdot \left(\prod_{j=1}^k \frac{2j}{2j+1} \right) = \frac{2(k+1)}{2(k+1)+1} \cdot \left(\prod_{j=1}^k \frac{2j}{2j+1} \right) = \prod_{j=1}^{k+1} \frac{2j}{2j+1}$$

The induction holds.

(iii)

The formula is $I_{2n} = \frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} \cdot \frac{\pi}{2}$.

Base Case: $n = 1$

$$I_2 = \int_0^{\frac{\pi}{2}} \sin^2 x \, dx = \int_0^{\frac{\pi}{2}} \frac{1 - \cos(2x)}{2} \, dx = \frac{1}{2} \left[x - \frac{\sin(2x)}{2} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$

The base case holds.

Inductive Step:

Assume the formula holds for some $k \geq 1$:

$$I_{2k} = \frac{\pi}{2} \prod_{j=1}^k \frac{2j-1}{2j} \quad (**)$$

We need to show the formula holds for $n = k + 1$, i.e.,

$$I_{2(k+1)} = I_{2k+2} = \frac{\pi}{2} \prod_{j=1}^{k+1} \frac{2j-1}{2j} = \frac{\pi}{2} \left(\prod_{j=1}^k \frac{2j-1}{2j} \right) \cdot \frac{2(k+1)-1}{2(k+1)}$$

Using the reduction formula from part (i) with $n = 2k + 2$:

$$I_{2k+2} = \frac{(2k+2)-1}{2k+2} I_{(2k+2)-2} = \frac{2k+1}{2k+2} I_{2k}$$

Substitute the inductive hypothesis (**):

$$I_{2k+2} = \frac{2k+1}{2k+2} \cdot \left(\frac{\pi}{2} \prod_{j=1}^k \frac{2j-1}{2j} \right) = \frac{\pi}{2} \left(\prod_{j=1}^k \frac{2j-1}{2j} \right) \cdot \frac{2k+1}{2k+2} = \frac{\pi}{2} \prod_{j=1}^{k+1} \frac{2j-1}{2j}$$

The induction holds.

Exercise 6.4 (30 pts)

We use the trigonometric identities:

$$\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

(i)

$$\int_{-\pi}^{\pi} \sin mx \cos nx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\sin((m-n)x) + \sin((m+n)x)] \, dx = \frac{1}{2} [-\cos((m-n)x) - \cos((m+n)x)]_{-\pi}^{\pi} = 0$$

(ii)

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\cos((m-n)x) - \cos((m+n)x)] \, dx$$

Case 1: $m \neq n$

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \frac{1}{2} \left[\int_{-\pi}^{\pi} \cos((m-n)x) \, dx - \int_{-\pi}^{\pi} \cos((m+n)x) \, dx \right] = 0$$

Case 2: $m = n$

$$\int_{-\pi}^{\pi} \sin^2 mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\cos(0) - \cos((2m)x)] \, dx = \frac{1}{2} \left[\int_{-\pi}^{\pi} 1 \, dx - \int_{-\pi}^{\pi} \cos(2mx) \, dx \right] = \pi$$

(iii)

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\cos((m-n)x) + \cos((m+n)x)] \, dx$$

Case 1: $m \neq n$

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = \frac{1}{2} \left[\int_{-\pi}^{\pi} \cos((m-n)x) \, dx + \int_{-\pi}^{\pi} \cos((m+n)x) \, dx \right] = 0$$

Case 2: $m = n$

$$\int_{-\pi}^{\pi} \cos^2 mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\cos(0) + \cos((2m)x)] \, dx = \pi$$

Exercise 6.5 (10 pts)

Substitute the series definition of $f(x)$ into the integral:

$$\int_{-\pi}^{\pi} f(x) \sin mx \, dx = \int_{-\pi}^{\pi} \left(\sum_{n=1}^N a_n \sin nx \right) \sin mx \, dx$$

Interchange the integral and the finite sum:

$$\int_{-\pi}^{\pi} \left(\sum_{n=1}^N a_n \sin nx \right) \sin mx \, dx = \sum_{n=1}^N a_n \int_{-\pi}^{\pi} \sin nx \sin mx \, dx$$

From Exercise 6.4 (ii),

$$\int_{-\pi}^{\pi} \sin nx \sin mx \, dx = \begin{cases} 0, & \text{if } m \neq n \\ \pi, & \text{if } m = n \end{cases}$$

Therefore, only the term where $n = m$ is non-zero in the summation.

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \sin mx \, dx &= a_m \int_{-\pi}^{\pi} \sin mx \sin mx \, dx + \sum_{n=1, n \neq m}^N a_n \cdot 0 = a_m \cdot \pi \\ a_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin mx \, dx \end{aligned}$$

Exercise 6.6 (10 pts)

$$E(P) = \frac{\lambda b}{4\pi\epsilon_0} \int_0^L \frac{dx}{[(x-a)^2 + b^2]^{3/2}}$$

Use trigonometric substitution. Let $u = x - a$. Then $du = dx$.

The limits change: $x = 0 \implies u = -a$ and $x = L \implies u = L - a$.

$$E(P) = \frac{\lambda b}{4\pi\epsilon_0} \int_{-a}^{L-a} \frac{du}{[u^2 + b^2]^{3/2}}$$

Now, let $u = b \tan \theta$. Then $du = b \sec^2 \theta \, d\theta$. $u^2 + b^2 = b^2 \tan^2 \theta + b^2 = b^2 \sec^2 \theta$.

Thus, $[u^2 + b^2]^{3/2} = (b^2 \sec^2 \theta)^{3/2} = b^3 \sec^3 \theta$.

The integral is:

$$\int \frac{b \sec^2 \theta \, d\theta}{b^3 \sec^3 \theta} = \int \frac{1}{b^2} \frac{1}{\sec \theta} \, d\theta = \frac{1}{b^2} \int \cos \theta \, d\theta = \frac{1}{b^2} (\sin \theta + c)$$

$\sin \theta = \frac{u}{\sqrt{u^2 + b^2}}$. Substitute back for u :

$$\int \frac{du}{[u^2 + b^2]^{3/2}} = \frac{1}{b^2} \left(\frac{u}{\sqrt{u^2 + b^2}} + c \right)$$

Now, apply the limits of integration from $u = -a$ to $u = L - a$:

$$E(P) = \frac{\lambda b}{4\pi\epsilon_0} \left[\frac{1}{b^2} \frac{u}{\sqrt{u^2 + b^2}} \right]_{-a}^{L-a} = \frac{\lambda}{4\pi\epsilon_0 b} \left[\frac{L-a}{\sqrt{(L-a)^2 + b^2}} + \frac{a}{\sqrt{a^2 + b^2}} \right]$$

Exercise 6.7 (10 pts)

A quadratic function has the form $f(x) = ax^2 + bx + c$.

Given $f(0) = 1$, $c = 1$.

Thus, $f(x) = ax^2 + bx + 1$. The integral is $\int \frac{ax^2 + bx + 1}{x^2(x+1)^3} dx$.

$\frac{ax^2 + bx + 1}{x^2(x+1)^3}$ can be written as $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3}$

By partial fraction:

$$E = \left[\frac{ax^2 + bx + 1}{x^2} \right]_{x=-1} = a - b + 1$$

$$B = \left[\frac{ax^2 + bx + 1}{(x+1)^3} \right]_{x=0} = 1$$

For an integral of a rational function to be a rational function (i.e., contain no $\ln$ or $\arctan$ terms), the partial fraction decomposition must not contain terms of the form $\frac{K}{x-r}$ (where the power is 1). So $A = C = 0$.

$$\frac{D}{(x+1)^2} = \frac{ax^2 + bx + 1}{x^2(x+1)^3} - \frac{1}{x^2} - \frac{a-b+1}{(x+1)^3} = \frac{b-3-x}{x(x+1)^2}$$

To make D computable, $b-3=0$. So $f'(0) = b = 3$.

Exercise 6.8 (10 pts)

The partial fraction decomposition has the form:

$$\frac{1}{x^n(x-a)} = \frac{A}{x-a} + \frac{C_n}{x^n} + \frac{C_{n-1}}{x^{n-1}} + \dots + \frac{C_1}{x}$$

Step 1: Find the coefficient of $\frac{1}{x-a}$

$$A = \left[\frac{1}{x^n} \right]_{x=a} = \frac{1}{a^n}$$

Step 2: Subtract the resulting term and simplify what is left.

$$\frac{1}{x^n(x-a)} - \frac{A}{x-a} = \frac{a^n - x^n}{a^n x^n (x-a)}$$

Using the factorization $a^n - x^n = (a-x)(a^{n-1} + a^{n-2}x + \dots + ax^{n-2} + x^{n-1})$:

$$\frac{a^n - x^n}{a^n x^n (x-a)} = \frac{-(x-a)(a^{n-1} + a^{n-2}x + \dots + x^{n-1})}{a^n x^n (x-a)} = -\frac{a^{n-1} + a^{n-2}x + \dots + x^{n-1}}{a^n x^n}$$

Thus, the partial fraction decomposition is:

$$\frac{1}{x^n(x-a)} = \frac{1/a^n}{x-a} - \frac{1}{a^n x} - \frac{1}{a^{n-1}x^2} - \dots - \frac{1}{ax^n}$$

Exercise 6.9 (20 pts)

(i)

$$\int_0^\infty \frac{dx}{\sqrt{x}(1+x)} = \lim_{t \rightarrow 0^+} \lim_{s \rightarrow \infty} \int_t^s \frac{dx}{\sqrt{x}(1+x)}$$

Use the substitution $u = \sqrt{x}$. Then $u^2 = x$, and $2u \, du = dx$.

The integral transforms to:

$$\int \frac{2u \, du}{u(1+u^2)} = 2 \int \frac{du}{1+u^2} = 2 \arctan u + C = 2 \arctan \sqrt{x} + C$$

Now, apply the limits:

$$\int_0^\infty \frac{dx}{\sqrt{x}(1+x)} = [2 \arctan \sqrt{x}]_0^\infty = 2 \left(\frac{\pi}{2} - 0 \right) = \pi$$

(ii)

$$\int_2^\infty \frac{1}{x\sqrt{x^2-4}} dx = \lim_{t \rightarrow 2^+} \lim_{s \rightarrow \infty} \int_t^s \frac{1}{x\sqrt{x^2-4}} dx$$

Use the trigonometric substitution $x = 2 \sec \theta$. Then $dx = 2 \sec \theta \tan \theta \, d\theta$.

The integral transforms to:

$$\int \frac{2 \sec \theta \tan \theta \, d\theta}{(2 \sec \theta)(2 \tan \theta)} = \int \frac{2 \sec \theta \tan \theta}{4 \sec \theta \tan \theta} d\theta = \int \frac{1}{2} d\theta = \frac{1}{2} \theta + C$$

Since $x = 2 \sec \theta$, we have $\theta = \operatorname{arcsec}(x/2)$.

$$\int \frac{1}{x\sqrt{x^2-4}} dx = \frac{1}{2} \operatorname{arcsec} \left(\frac{x}{2} \right) + C$$

Now, apply the limits:

$$\int_2^\infty \frac{1}{x\sqrt{x^2-4}} dx = \left[\frac{1}{2} \operatorname{arcsec} \left(\frac{x}{2} \right) \right]_2^\infty = \frac{1}{2} \left(\lim_{s \rightarrow \infty} \operatorname{arcsec} \left(\frac{s}{2} \right) - \lim_{t \rightarrow 2^+} \operatorname{arcsec} \left(\frac{t}{2} \right) \right) = \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi}{4}$$

Exercise 6.10 (20 pts)

(i)

$$I = \int_0^\infty \frac{f(x)}{1+x^n} \frac{dx}{x}$$

Apply the substitution $x = \frac{1}{u}$. Then $dx = -\frac{1}{u^2} du$.

When $x = 0$, $u = \infty$. When $x = \infty$, $u = 0$.

$$I = \int_\infty^0 \frac{f(\frac{1}{u})}{1+(\frac{1}{u})^n} \frac{1}{\frac{1}{u}} \left(-\frac{1}{u^2} du \right) = \int_0^\infty \frac{f(\frac{1}{u})}{1+u^{-n}} \frac{1}{u} du$$

Use the given property $f(\frac{1}{u}) = f(u)$:

$$I = \int_0^\infty \frac{f(u)}{1+u^{-n}} \frac{1}{u} du$$

Then there is

$$I = \int_0^\infty \frac{f(x)}{1+x^n} \frac{dx}{x} = \int_0^\infty \frac{f(x)}{1+x^{-n}} \frac{dx}{x}$$

$$2I = \int_0^\infty \frac{f(x)}{1+x^n} \frac{dx}{x} + \int_0^\infty \frac{f(x)}{1+x^{-n}} \frac{dx}{x} = \int_0^\infty \frac{(2+x^n+x^{-n})f(x)}{(1+x^n)(1+x^{-n})} \frac{dx}{x} = \int_0^\infty f(x) \frac{dx}{x}$$

Thus $I = \frac{1}{2} \int_0^\infty f(x) \frac{dx}{x}$

(ii)

The integral is in the form of the LHS of part (i), where $f(x) = \frac{x}{1+x^2}$.

Therefore, use the conclusion derived in (i):

$$\int_0^\infty \frac{dx}{(1+x^n)(1+x^2)} = \frac{1}{2} \int_0^\infty \frac{dx}{1+x^2} = \frac{1}{2} [\arctan x]_0^\infty = \frac{\pi}{4}$$